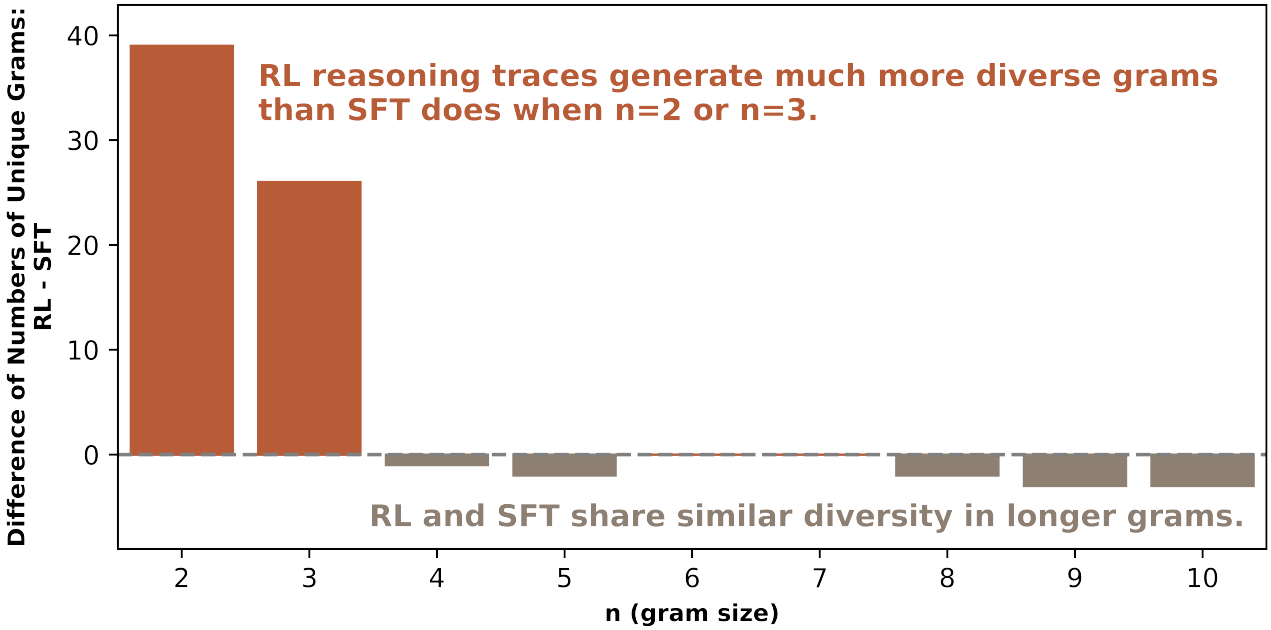

Figure H1. **Difference of numbers of unique grams between the RL and SFT model.** For the n-grams obtained in Appendix D (lines 979-983), we calculate the number of unique grams for both RL and SFT models, then visualize their differences here. The results show that (1) RL reasoning traces generate much more diverse grams than SFT does when $n = 2$ and $n = 3$; (2) RL and SFT reasoning traces share similar diversity in longer grams.

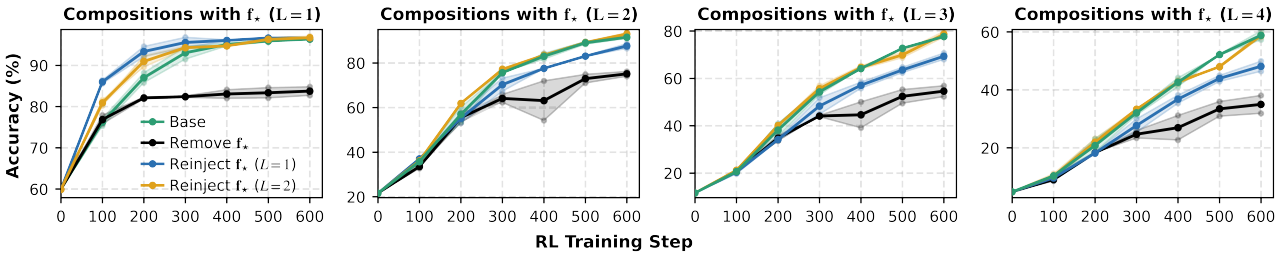

Figure H2. **The results for running partial experiments of Figure 5 in the initial submission across multiple seeds.** We follow all configurations of Figure 5 in the initial submission and only use a different training seed. The solid line represents the average result over two runs, while the shaded region shows the variation across runs. Our conclusions of Figure 5 remain the same since the noise induced by the seed parameter is limited.

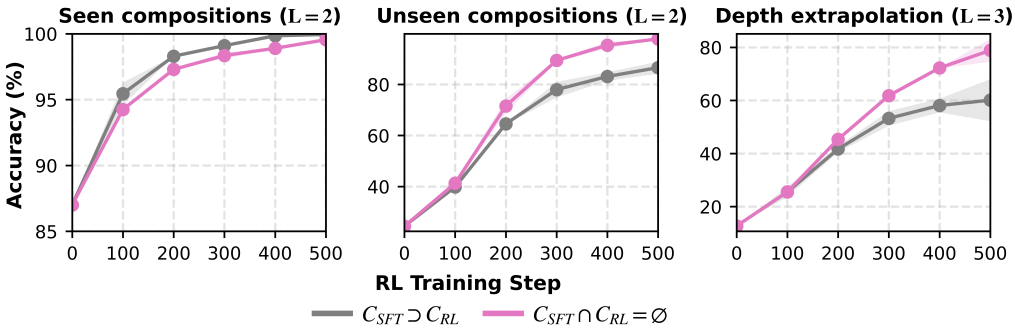

Figure H3. **The results for running partial experiments** ($L = 2$) **of Figure 7 in the initial submission across multiple seeds.** We follow all configurations of Figure 7 in the initial submission and only use a different training seed. The solid line represents the average result over two runs, while the shaded region shows the variation across runs. Our conclusions of Figure 7 remain the same since the noise induced by the seed parameter is limited.

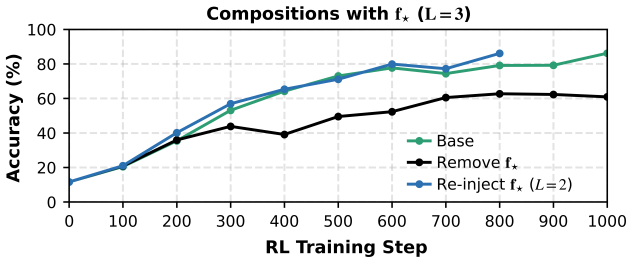

Figure H4. **The results for continually running partial experiments of Figure 5 in the initial submission to a more converged status.** As the training step increases, the curves gradually stabilize; the gap between the black line and the other curves does not diminish, indicating that the conclusion drawn in Figure 5 remains valid. Although the blue line doesn't reach 1,000 steps due to computational resource and time constraints, the current results still indicate that the blue and green lines are similar. We will provide the complete experimental results in the revision.

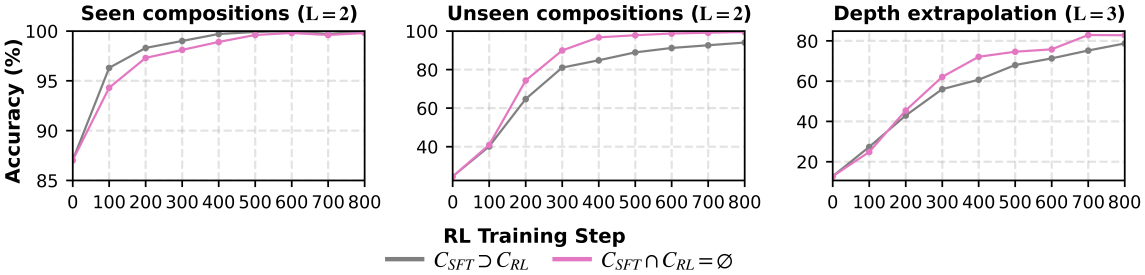

Figure H5. **The results for continually running partial experiments ($L = 2$) of Figure 7 in the initial submission to a more converged status.** As the curves approach a state of convergence, the gap between the two narrows. Nevertheless, the results indicate that the configuration represented by the pink line outperforms that of the gray line; the former accelerates the convergence rate, thereby achieving superior performance within a limited training time. This finding aligns with the conclusions drawn from Figure 7.